The Application of Mapping in Composition and Design

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Abstract

Mapping concerns the connection between structures, or gestures and audible results in a musical performance or composition. This is important to computer music composition as most of it is algorithmic in some way or another and it involves mapping. Algorithmic composition is sometimes the process of imagining a gesture or structure - perhaps physical or visual - and then applying a mapping process to turn that structure of the conceptual domain into sound which may display the original conception in some way. This article looks at mapping from the point of view of Australasian algorithmic composition practice, particularly where persistence is an issue, such that the structure (conceptual domain) is embodied and perceptible in the musical result. An attempt is then made to draw some parallel to the practice of (visual) design that uses mapping and examine the similarities and differences with algorithmic music composition.

1 Introduction

Mapping as part of algorithmic composition has not been as thoroughly investigated as mapping in instrument design. This is possibly because it is usually an integrated part of the composition process and not seen as a discrete stage of practice. Mapping in algorithmic composition is different from mapping in instrument design because composition is a process of planning and instruments are for real time music production. The use of mapping which is integral to an instrument has both differences and similarities to the mapping that is an inherent part of algorithmic composition. This paper investigates how some Australasian composers use mapping in their practice and moves towards a taxonomy of the process.

Another aspect of this investigation is how mapping is used by architects and designers. Some designers say that they explicitly use mapping in their practice. There is an attempt to analyse this in a small way and see if there are parallels with composition practice.

2 Defining The Terms

Compositional structures and ideas can take many forms, but they are often abstract in some way, to a greater or lesser degree, from the music that is composed. Composers sometimes use visual ideas of shapes, mathematical functions, physical processes, or phenomena and so on as ideas for creating music. Mapping is the process of taking the (possibly abstract) compositional structures and generating musical parameters. Gesture is an ill-defined term, which for the purposes of this paper means a musical concept; it is not a physical movement.

A musical gesture is a planned change (randomness can be planned) in musical parameters as part of a piece of music. The parameters could be, for example; timbre, density, intensity, timing, pitch, and so on. A compositional gesture is the underlying conception, structure and planning of the musical gesture.

"Conceptual domain" is a term that will be used to cover the entire conceptual area of compositional practice, which includes other organisation strategies as well as compositional gesture.

3 Mapping In Composition

Algorithmic composition can be the practice of composing music through manipulating a number of compositional concepts, structures or gestures to produce a musical piece. The manipulation is usually part of the compositional process, where a multitude

of gestures or structures are arranged and then through another process, possibly very direct, musical parameters are extracted. This is most usually done with the aid of a computer, although it should be noted that this is usually a matter of efficiency and expediency only. It is possible to do this from the micro-composition scale of sound synthesis (where it can only be done with a computer) through to the macro scale of the structural model of the piece. Mapping is the process of extracting the musical parameters from the compositional gestures or structure, such that one set of data is *mapped* onto another. It should also be noted that algorithmic composition might use specific organisations of data that do not necessarily require "gestures"; for example, Markov chains, finite-state machines, fractals and space grammars. These constructs may also require significant mapping processes to output musical data.

There may appear to be some similarities between the methods of mapping in algorithmic composition and the mappings used in the general practice of the sonification of data. This can sometimes be a valid observation with some pieces of music, and there is certainly some overlap between the two areas. However, as the intent of sonification is to illustrate a process or some data, it is primarily instructional. This type of mapping is not the focus of this paper, which is primarily about *composition* and mapping, which is somewhat more ephemeral. The closest sonification comes to composition, what Larry Polansky calls "manifestation" (Polansky 2002), is where the intent is to creatively use a type of sonification of a formal or mathematical process to create a musical idea. For the purposes of this paper, that process will be included as a part of composition.

4 Historical Examples

Mapping has been used by some composers for many years, although it may not have been identified as a separate element of practice. These historical examples will help to explain the function of mapping, showing how it has been used by composers.

A famous example of the mapping process is the part of *Pithoprakta* (1955-56) as reproduced in Formalized Music pp17-21 (Xenakis 1991). Xenakis used the Brownian motion of gas particles, combined with Bernoulli's Law of Large Numbers, as his basic model for the cloud pizzicato-glissandi section. After calculating, statistically, over 1000 velocities of gas particles at given instants of time (the measurement of this was impossible), he then graphed them on an XY plane and directly mapped the straight lines of the velocities to glissandi for 46 string instruments. Xenakis divided his graph vertically into 15 pitched sections, each corresponding to a major third. This was then mapped to the ranges of the string instruments. The mapping was directly of pitch in the vertical direction. This is a particularly direct and concrete example of mapping. All intensities and durations are the same, but to ensure the sensation of a cloud of particles, Xenakis used a complex temporal arrangement of overlapping timing subdivisions that are factorially-unrelated. This is a complex mapping of linear time, designed to represent the instantaneous nature of the movement of the gas particles. Along with many other algorithmic composers, the mapping phase is implicit in much of Xenakis' work. He often used direct mapping as a result of the deliberate organisation of one set of data in such a way that it maps directly to musical parameters.

Another famous example of algorithmic composition and mapping is Charles Dodge's *Earth's Magnetic Field* (1970). Here, Dodge used data from the effects of the radiation of the sun on the magnetic field of Earth. A Bartels diagram showed fluctuations in the Earth's magnetic field for 1961 and this data formed the basis for the piece. Dodge mapped this data, the Kp index (a measure of the average magnetic activity of the earth) to pitches and rhythms. From the program notes of the recording of *Earth's Magnetic Field* (Dodge, 1970), we may glean an insight into the mapping used:

"The succession of notes in the music corresponds to the natural succession of the Kp indices for the year 1961. The musical interpretation consists of setting up a correlation between the level of the Kp reading and the pitch of the note (in a diatonic collection over four octaves), and compressing the 2,920 readings for the year into just over eight minutes of musical time." (Dodge, 1970)

While the pitches appear to be a direct mapping from the Kp index, some elements of the composition such as the timbres were chosen purely for aesthetic effect. An arrangement of the data that plotted the

length of sections of the data against the maximum amplitude in the section was used to determine the speed and direction of the sound spatialisation and the rhythms. The data was also sometimes read multiple times to generate the musical parameters. That, combined with the similarity of the fluctuations in the Kp index to 1/f noise data, contributes to the aspects of self-similarity in the piece.

The two previous examples contrast different approaches to mapping. It may be linear and direct, but it may also be nonlinear and more complex. Both examples use the data as a structural component and the music achieves some structural unity for that. Also, both of these examples come from a time when computer music practice was in its infancy. With current tools and thinking, composers can use mapping in much more creative and indirect ways.

5 How Mapping Is Used

There has not been the same formalisation of algorithmic composition and mapping as there has been for other musical composition. Additionally, there is little formal analysis of music in terms of mapping or compositional gesture. As there is no set method for defining the mappings or structures, each composer tends to use their own methods for their own reasons.

Compositional structures and mappings are also used differently by different composers, thus making this a problematic area for analysis and study: For example, the structural model and mapping for a piece of music will have different applications for a composer who is focused on, say, spectral composition in contrast to another focused on some other form of algorithmic composition.

It is clear from descriptions of how composers use mapping between the conceptual or abstract domain and the musical domain, that it is sometimes used in a similar combination of ways to how it is used in instrument design. That is; mapping one compositional parameter to many musical parameters (one to many), mapping many compositional parameters to one musical parameter (many to one) and mapping many compositional parameters to many musical parameters (many to many). Additionally, there is another parallel between instrument designers and composers, the mappings themselves may be linear mappings or nonlinear mappings (Hunt and Wanderley 2000, Hunt, Wanderley and Kirk 2000). However, in compositional mapping there is the additional possibility of repetitive nonlinear mappings and most importantly, each composer has their own combination of mapping techniques.

From the comments made by the architectural designers questioned, it seems that there is some overlap in the use of mapping to move from the abstract to the concrete. This is an area that is only touched upon here, to demonstrate that this is (yet another) intersection between music and architectural design.

6 Questions Of Mapping

To understand how Australasian composers use mapping, a number of them were asked questions concerning their practice. The broad answers are summarised below. The composers were; Roger Alsop, Rodney Berry, Chris Cree Brown, Phil Brownlee, Warren Burt, Densil Cabrera, Tim Kreger, Peter McIlwain, Gordon Monroe, Garth Paine, Greg Schiemer, and Paul Doornbusch. The five designers questioned were Pia Ednie-Brown, Tom Kovak, Paul Minifie, Vivian Mitsogianni and Julian Raxworthy. For the designers the questions were modified to exchange “architectural design” for “composition”, “viewer” for “listener” and “visual output” for “musical output”, and so on. The questions, asked were:

1. Is mapping something that you are conscious of when you are composing?

2. Do you have a consistent approach to mapping and (algorithmic) composition or does it vary and why?

3. When implementing a mapping strategy for (part of) a composition, do you organise this in a particular ‘analytical’ way (decomposing the problem in a technical manner), or in a more creative and holistic way for a purely aesthetic result?

4. Is the mapping component of your compositions something that might be perceptible by a listener, or of interest to them and why?

5. Is the mapping component of algorithmic composition something that is pre-determined for you or is it part of a process of exploration?

6. Do you use individual mapping strategies for individual parameters or is there reuse of mapping

strategies or a global system? (ie are they monoparametric or multiparametric?)

7. What elements do you control algorithmically in a composition and can you comment on the function and importance of these?

8. Is the mapping consistent within these elements or not?

9. Are the mapping schemas you use mostly linear mappings or non-linear in some fashion, and why?

10. Can mapping be considered a composition technique in itself?

There was deliberate overlap in the questions to elicit as much information as possible and the respondents were also asked to make a yes/partly/no, multiple-choice response to each question. These questions will be expanded upon later.

7 Responses

The results of the composers' responses are summarised below, making it possible to note similarities and divergences. The answers supplied by the composers are in the appendix to this paper. In this summary, it is possible to achieve the start of an overview of this area.

7.1 General remarks by respondents:

While composers may use mapping to move from the conceptual domain to musical parameters, mapping can work backwards for listeners, providing access to structures and concepts from the music. Composers tend to use complex ideas and structures for music. Focusing on mapping assumes that something pre-exists before the act of composition and not all composers may think or work like that. If mapping is a means of moving from the conceptual domain to the concrete domain, then it may be important for all composition, but it is central to algorithmic composition.

7.2 Collated Responses:

The tables below show the general response versus the number of composers (C) or designers (D) who had that response. While the number of people questioned means that this is a statistically minuscule sample, it does point to trends and variations.

Question 1.	C	D
General Response	#	#
Yes, I am conscious of using mapping	8	3
Sometimes I am conscious of using mapping	3	2
No, I never consciously use mapping	0	0

Question 2.	C	D
General Response	#	#
I have a consistent approach to mapping	0	0
I sometimes have a consistent approach to mapping	5	3
I do not have a consistent approach to mapping	6	2

Question 3.	C	D
General Response	#	#
I mostly use mapping analytically	0	1
Mapping is sometimes used analytically	3	2
I mostly use mapping creatively	6	2

Question 4.	C	D
General Response	#	#
Mapping is of great interesting for listeners	2	1
Sometimes mapping is interesting to listeners	9	4
Mapping is not interesting to listeners	0	0

Question 5.	C	D
General Response	#	#
Mapping is predetermined in composition	1	0
Mapping is sometimes predetermined, sometimes part of exploration	4	2
Mapping is part of the exploration of composition	6	3

Question 6.	C	D
General Response	#	#
I use individual mapping strategies for individual parameters (monoparametric)	2	0
Both individual mappings strategies are used as well as reusing some	7	2
I reuse mapping strategies for multiple parameters (multiparametric)	1	3

Question 7.	C	D
General Response	#	#
I usually control as many elements as possible	2	1
I control most elements sometimes, fewer at other times	4	0
I usually control some elements only	5	2

Question 8.	C	D
General Response	#	#
The mapping is mostly consistent for each element	9	3
The mapping is sometimes consistent for each element	1	0
The mapping is mostly not consistent for each element	1	2

Question 9.	C	D
General Response	#	#
Mostly linear mappings are used (very complex structures)	1	1
A mixture of linear and nonlinear mappings are used (depends on structure)	7	1
Mostly nonlinear mappings are used (simpler structures)	3	3

Question 10.	C	D
General Response	#	#
Mapping can be a compositional technique in itself	6	3
Mapping is part of compositional technique	5	2
Mapping cannot be considered a compositional technique at all	0	0

8 Types Of Mapping

From the composers' descriptions of how they use mapping, it seems that there is no set method for mapping data from the domain of the conceptual, gestural or structural to the musical domain. Linear and simple mappings are *sometimes* used by composers, but it is used significantly less often than mappings that are more complex.

Simple mappings have been most clearly described as ratiometric (Polansky 2002). In this type of mapping, a doubling of the data to be mapped results in a "doubling" of the musical parameter. This could be, for example, an octave pitch displacement or a doubling of the loudness. Note that this second linear mapping might be perceptually linear to a particular person under particular conditions, but it would be mathematically nonlinear. Because ratiometric mappings are the easiest to perceive they are particularly useful when the composer has some data, gesture or concept that should be translated as directly as possible for the listener. The previous example of the pitch mappings from Xenakis' Pithoprakta is an illustration of this. This particular example is musically successful because the data to be mapped is very complex from the outset. Simple mappings can be less musically successful if the data to be mapped is very simple.

Complex mappings may be regularly nonlinear, such as an exponential law. For example, a square law will produce a fourfold increase in the musical parameter from a doubling of the data to be mapped. If this is perceptually based it can be perceived as a more extreme or less extreme mapping with changes in the data to be mapped. Other types of complex mappings are as varied as can be imagined. They can sometimes be related to a complex arrangement of the data to be mapped, or other potentially chaotic functions can be involved. This may obscure the original concept, compositional gesture, or data, or it may embellish it and give it another dimension. However, there is clearly a limit to how far such complexity in mapping can be taken before all sense of the original data is lost in the mapping. Polansky convincingly suggests that the cognitive weight of complex mappings degenerates rapidly and nonlinearly such that beyond a certain point everything is just "complex".

It has been a repeated outcome from instrument design research that humans prefer complex mappings (Hunt and Kirk 2000). This would appear to translate somewhat into the domain of composition, given the previous situation that linear mappings may be most appropriate for very complex data.

In one way, the variety of sophisticated, creative and exploratory approaches to mapping, as embodied in the works of algorithmic composers, is a wonderful outcome and it ensures the variety of music from algorithmic composition. In another way, it means that mapping strategies are reinvented by every composer and young composers do not have easily accessible models to work from. If there was created a categorisation of compositional mapping strategies and a catalogue was built of mapping techniques, then other composers could refer to this and build on it in their own practice. This would be of particular use to student composers (Polansky 2002). In this way, mapping in algorithmic composition could be demystified and more complex, varied and musically appropriate practices could be developed by building on the work of others. The counter argument to this is that there is something valuable in the effort of developing sophisticated mappings for oneself. In instrument design there is certainly interest in the

effort required to play an instrument and how mapping relates to this (Ryan 1992).

The answers from the designers are interesting for a number of reasons. There appears to be some degree of overlap in the way that composers use mapping and how architectural designers use it. This is an additional parallel than those that are usually drawn between architecture and music; that they both communicate through, and are deeply concerned with, the concept of form, and that one divides space while the other divides time. The similarities in the ways both designers and composers use mapping, and also how they do not use mapping (the 0 scores in the tables above), points to some similarity in the creative process between the disciplines. However, I think there is both too much semantic variation in the understanding of the questions presented, and too few people involved, to make anything more than a general claim of similarity.

Something which has not been explored here, but which would be of interest to some composers, are cultural associations with mapping. For example, are there culturally invariant mappings with some parameters such as pitch or intensity with height? One example from instrument design that has caused problems is the original Theremin mapping of intensity with proximity to a horizontal antenna such that to make a louder sound the performer would bring their hand *down* to the volume antenna. This was done for practical reasons so that without a performer there is no sound, but modern theremins allow this to be swapped because of the cognitive dissonance caused by the playing action.

9 Conclusions

One clear outcome of this is that the questions need improving if this is going to be attempted again. This is indicated clearly in the detailed responses where not all of the questions seemed sensible to some people, but it is also evident in the spread of answers to some of the questions, although some spread is to be expected of course. In addition, the semantic understanding of many of the terms surrounding mapping and composition (and design) seems to be ill defined and this leads to a varied understanding of the questions and therefore variable answers.

Not withstanding the problems with the questions and semantics, some valuable information can be extracted from the answers. The answers indicate that composers who use algorithmic techniques certainly use mapping from one domain to another, often in very sophisticated ways. Also indicated is that there is no ideal solution or single method, but that it is largely a process of exploration. Very simple or obvious mappings are sometimes problematic in composition probably because of the oversimplification of the resultant music, but they may be appropriate under some circumstances such as with very complex structures or data. More complex mapping strategies are more common and more musically useful, but very numerous and their application and musical usefulness appears to be an aesthetic judgement of the composer. Composers appear to be aware of complexity that does not scale linearly for listeners such that there is some limit to what makes sense to a listener's perceptions and beyond this it is so complex that there is no sense of the original concept or data. There seems to be almost as many approaches to mapping as there are composers or designers. It is the character of composition, being an expansive and creative activity, that if ever a theory provided a "standard", "optimised" or "ideal" mapping technique or repertoire, that composers would ignore it and go beyond that. Having said that, there would appear to be *something* to gain by creating a catalogue of historical and current mapping practices, even if only in a pedagogical sense (Polansky 2002) and this could be started by expanding on the questions presented above.

The nature of composition and composers means that there will never be "solution" to the mapping problem in algorithmic composition, that it will remain a part of the technique and exploration of composers.

Not being a designer, I am unqualified to comment on the area of design, but the survey showed significant areas of similarity in the responses. Significant enough, I feel, that there is some similarity of practice and theoretical approach such that further investigation and cross-discipline discourse is warranted to inform both disciplines.

10 Acknowledgments

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References

Wanderley, M. M. and Battier, M. eds. 2000. *Trends in Gestural Control of Music*. Paris: IRCAM

Cadoz, C. and Wanderley, M. M. 1999. *Gesture-Music*. In Trends in Gestural Control of Music. Paris: IRCAM CDROM.

Dodge, C. 1970. *Earth's Magnetic Field program notes*. In Nonesuch LP H 71250 EARTH'S MAGNETIC FIELD: Realizations in Electronic Sound.

Dodge, C. and Jerse, T. 1997. *Computer Music second edition*. New York: Schirmer Books

Hunt, A., Wanderley, M. M. and Kirk, R. 2000. *Towards a Model for Instrumental Mapping in Expert Musical Interaction*. In Proceedings of the 2000 International Computer Music Conference. San Francisco: International Computer Music Association, pp. 209 - 212.

Hunt, A. and Wanderley, M. M. (eds.) 2000. *Mapping of Control Variables to Musical Variables*. Interactive Systems and Instrument Design in Music Working Group. Website: http://www.notam02.no/icma/interactivesystems/mapping.html

Hunt, A. and Kirk, R. 2000. *Trends in Gestural Control of Music*. Paris: IRCAM

Kramer, G. 1996. *Mapping a single data stream to multiple auditory variables: A subjective approach to creating a compelling design*. In Proceedings of International Conference on Auditory Display - ICAD'96.

Matossian, N. 1986. *Xenakis*. London: Kahn & Averill

Polansky, L. 2002. *Manifestation and Sonification*. Unpublished paper.

Roads, C. 1996. *The Computer Music Tutorial*. Cambridge, Massachusettes: The MIT Press

Ryan, J. 1992. *Effort and Expression*. In Proceedings of the 1992 International Computer Music Conference. San Francisco: International Computer Music Association, pp. 414 - 416.

Thieberger, E. M. 1995. *An Interview with Charles Dodge*. Computer Music Journal 19(1): 11-25

Varga, B. A. 1996. *Conversations with Iannis Xenakis*. London: Faber and Faber

Xenakis, I. 1991. *Formalized Music Pendragon Revised Edition*. New York: Pendragon Press

Appendix 1: Detailed Responses

Below are the detailed comments from the composers and designers. The various approaches to mapping in algorithmic composition and also design can be easily seen in the responses. Additionally, the comments by the designers show the overlap and differences with composers in thinking about mapping. These comments have been very lightly edited for readability only; I have added no words of explanation.

The composers questioned were; Roger Alsop (**RA**), Rodney Berry (**RB**), Chris Cree Brown (**CCB**), Phil Brownlee (**PB**), Warren Burt (**WB**), Densil Cabrera (**DC**), Tim Kreger (**TK**), Peter McIlwain (**PM**), Gordon Monroe (**GM**), Garth Paine (**GP**), Greg Schiemer (**GS**), and Paul Doornbusch (**PD**). The designers questioned were Pia Ednie-Brown (**PEB**), Tom Kovak (**TXK**), Paul Minifie (**PXM**), Vivian Mitsogianni (**VM**) and Julian Raxworthy (**JR**).

1. Is mapping something that you are conscious of when you are composing?

PM: In the sense that mapping is some process that transforms material from state to another then all standard compositional procedures (such as transposition, inversion etc) are a form of mapping. I am not always conscious of these procedures as they have become more automatic over time. However, the term mapping is often used in a specific way to refer to transformations of musical materials in correlation with extra-music materials. When I have done this I am usually very aware of the process chiefly because I feel that must make judgments about the effectiveness in musical terms (than is, my own subjective set of musical definitions) of this, or that, mapping procedure.

WB: If the piece I'm doing involves the application of some sort of numerical data, whether a simple use of a random number generator, or something more complex, I'm concerned with mapping. Numerical sources only assume meaning when they're applied to materials that have a kind of "graspability" to them. That is, to hear a number pattern for its own sake, it might be better at first to apply it to a set of 5 pitches rather than all 12 (if we're talking 12tone equal temperament for the moment), because with fewer pitches we might be able to "forget" the traditional musical implications of the fewer pitches (ie, they're so bland we can forget them for the moment) and concentrate on the qualities of our numerical structure. When it comes time to put those structures into a larger "piece", then a much more conscious selection of materials can take place.

GM: Most, but not all, of my compositions involve mapping in some way.

DC: It is not a term that I would normally apply. Nevertheless, I'm conscious of the ideas of 'transcription' or 'sonification', which seem to be closely related - if I understand what you mean. In some cases, my works have realised the acoustic implications of a spatial journey, and I suppose the term 'mapping' is especially apt there.

GP: When composing a piece of music, or a soundscape, I view the mapping of the sound material to the intended emotional context of the work to be central to the craft of composition. I am usually contemplating at least two layers of material at any one time - an environment/broader context, and individual lines of activity/dialogue/narrative.

PD: Yes, I'm conscious of mapping when I'm working, but it's not the first thing I think of. I'm very conscious of it when I'm working on transcribing a structure or gesture into music or sound. However, I am not sure if I think that it is important, in my work, that the compositional structures or gestures are somehow transmitted to the audience. In a way it is, but mostly I'm trying to give the audience a powerful musical experience, so as far as transmitting a gesture or structure is important for that, then yes it is important. I try to map from compositional gestures and structures to musical parameters in such a way that the gesture or structure is evident when *I* listen to the result. This may or may not translate to these being perceived by the audience, but I must be able to hear it. Having said that, most of the compositional gestures and structures I use are abstract concepts and I usually have movement in three or four of them simultaneously, otherwise I find the result boring musically.

PEB: Mapping is always an integral part of architectural design. All aspects of design involve representational devices (plans, perspectives, models etc). The act of moving between various modes of representation and between drawings and building

always involves many types or modes of mapping. As such, mapping is integral and inevitable; some aspects of which are more conscious than others.

The question gathers a more elaborate (and perhaps more precise) answer for me when I think of the sculptural work I've done. In this case the outcome is the object itself, there is not an imagined future event of actualisation (building), making what I see as the primary nature of mapping very directly accessible. I made a series of casts that emerged through experiments with latex and plaster. Each one of the series is the product of a different deformation of the same mould. They were made through pouring liquid plaster into the latex mould of a shower tap. The act of making these involved a kind of puppeteering act in which the stretchy, elastic moulds were suspended with string, positioned with wire and held down with tape under chairs, off coat hangers - whatever was at hand. The latex stretched and deformed with the weight of the fluid plaster, particularly as the makeshift supports fell away or shifted. Each casting was enacted with a different placement or 'context' of the mould. Although the process could be guided toward expected results, the final cast was always a surprise, being very contingent to the moment. The resulting series of different casts arose from the coalescing involutions of a myriad of variable relations (ie the outcome is a mapping of the complexity that is involved in the situation - it is a qualitative mapping of an event) all struck into an accord: form extracted from dynamic interactions, falling into one another in a collaborative agreement with a responsively overarching skin. In this example, it is a *situation* that is explicitly mapped more than anything else. I would argue that this is *always* the case in a creative act, no matter what the process, even if the dynamic variability of the moment or situation is embedded (and rendered opaque) within a rigid, linear, direct mapping process. Often, where extremely non-linear mapping positions are taken, there is an assumption that the material being mapped is the subjectivity of the author. What I like about the plaster cast example is that, while being manifestly non-linear, like many programmed acts of mapping, there is an implied shift in the role of the author/creator in that the outcomes are, to a relatively explicit extent, a product of the process (where designing the process itself becomes the primary role of the composer).

PXM: In some projects mapping is of crucial interest - it is a key way of capturing relationships and structures into a design work. It helps because it is an antitectonic strategy, disrupting a normal tectonic set of relations.

VM: My own work deals with a 'machine based' design methodology, that is the process is designed rather than the object as a starting point. The 'process' or 'machine' will often include mapping, diagramming in part. This is a generalisation, but the work generally involves the use of techniques that contain abstract information (such as diagrams, scientific systems of behaviour) and are used as part of the larger design process. This is more akin to a 'machine' based design process in the sense of Deleuze's use of the term 'machine' in that the 'mapping' is a part of the machine that (along with other ideas and techniques) are 'applied' to something, they are not the thing itself. That is, mapping is not the main tool used.

2. Do you have a consistent approach to mapping and (algorithmic) composition or does it vary and why?

TK: Each algorithm or process I've used exhibits different behaviour and hence requires different treatment in order to obtain sonically or musically meaningful and interesting results.

PM: For me the use of mapping (in relation to extra-musical material) usually occurs when I am experimenting. Therefore, I will tend to explore a more diverse range of options. However I have noticed that over time I have tended towards the use of extra-musical processes, such as flocking or neural network behaviour, as a means of musical parameter generation instead of a more empirical approach where an extra-musical data set is mapped to onto a musical parameter set. This latter approach I feel is a little too arbitrary and difficult to control effectively in a musical context.

WB: When I'm testing out a new microtonal scale, I'll usually just have an equal-weighted random number generator pumping out an octave or two of the scale with no dynamic variation and just a simple pulse rhythm, so I can get the feel for the harmonic world of the scale. However, when I sit down to compose a more elaborate thing, then I'm thinking about which parameters should be controlled in which way, why, and how. And that can vary with

each piece. There does seem to be a sort of divide in my composing - if I'm controlling "notes" - or what behave like traditional notes - then I seem to think very carefully about both the kinds of patterns I'm using, and the materials they're applied to. But if I'm controlling "samples" - that is, fragments of real-world sound that don't behave like traditional notes (examples - Ruth Westheimer saying "pre-marital sex", or a recording of the great hall in Union Station, Chicago), then my approach to mapping is more determined by the sonic nature of the sample, and the overall effect I want the sample to have.

GS: Mapping creates the connections between the parameters in a musical process and how it is perceived by the listener/performer.

DC: The details of the approaches vary between compositions. But I have a consistent attitude to compositional work, which is that I never wish to control the details of the sound. Instead, I like to make a very few large scale decisions, from which the sound is discovered. In recent years I have also found an interest in merely recording sound - I guess this is a bit different to the algorithmic stuff I've done. And several pieces are really about physical objects, so it is the object that determines the range of sound possibilities. Within a work, the approach is consistent. Between works there is some variation. Maybe some works don't involve mapping within the boundaries of your definition.

GP: Mapping is different for each work. In my installation practice there are some approaches to mapping which I have been trying to develop. These focus on dynamic mapping, where the mapping regimes change in accordance with the dynamic of movement sensed in the space. The gesture to synth. Variable relationships change and so does the or orchestration. The orchestration of these works is developed on an object-oriented programming basis so that the synthesis resources change in accordance with changes in gesture dynamic and recognised patterns of movement. The only consistency in my composition practice is my approach to an environment/broader context, and individual lines of activity/dialogue/narrative as discussed above.

PD: I think I would have to say that I do not have a consistent approach to the mapping of compositional gestures to musical parameters. It will certainly change from piece to piece. I have recently used linear, logarithmic and chaotic mappings for time, instrumental pitch, synthesis pitch, timbre, density and other parameters in a piece. I prefer this approach as I find it produces a more interesting result than using a single technique. This may be particular to the types of models I use, for instance when I have had very complex data to map to musical parameters I usually find that a more linear mapping allows for a better result, than a non-linear mapping, giving more sense of the original idea.

PEB: I'd say it's consistent (if a variable consistency) but only in the sense that, no matter what the parameters, there is an attempt to produce a consistency - or in other words, a coherent quality (like the soft and fluid but not too thin and runny yolk of a boiled egg). The coherent quality of anyone's work is what we generally see as their 'style', as that which makes it recognisable. As such, anyone who has managed to produce a recognisable style has a consistent approach to mapping even if the details and processes involved vary wildly.

JR: Design methodologies are adapted to suit the specific nature of the brief / interest / situation. Methods are hybridised.

PXM: I have a mixed relationship with trying to achieve a consistent and complete representation of a structure from elsewhere in design. One significant desire is to achieve a perfect reflection of a structure within the project - to the point where the inverse operation could potentially recreate the original structure from the work. Resolution is one problem in design, where contingencies and constraints add inevitable noise to the operation. So I like to keep projects in a kind of proto - or pre – form, where the operations are most clear.

However taking things towards resolution usually requires that the operation does not have such complete control over the design - then the operation is used to call up a concept - and its expression is more a representation of a structure and its potential embedding within objects. To some extent this is an aesthetic choice as well as a pragmatic one – it seems proper to acknowledge the contingent and partial nature of designed things, and to let the structure exist more as an idea within a work rather allowing it total governance.

VM: I have a relatively consistent approach, that alters according to a large number of factors, (ie. the other issues involved in the design process). Trial & error also (ie. the mapping is set up in the form of a

pseudo scientific experiment) this after the 'results' are observed, often takes the project in different direction.

3. When implementing a mapping strategy for (part of) a composition, do you organise this in a particular 'analytical' way (decomposing the problem in a technical manner), or in a more creative and holistic way for a purely aesthetic result?

TK: Ultimately my main concern is for the integrity of the algorithm or process, although aesthetic is always the issues. I never bend an algorithm for the sake of the aesthetic nor do I compromise on the aesthetic for the sake of the algorithm, the two play equal and integral parts to the entire piece.

PM: I find that as the work usually occurs on the computer that it is easier to experiment first and then go into the detail and analyse what is happening. I guess this is because I tend to work with processes that can be difficult to predict exactly what will happen. Usually I have a sound or texture in my mind that I want to achieve and then change the procedure until it generates the kind of result that I was looking for.

WB: I sometimes use different processes for different parameters. I might, for example, use the x-axis of a particular fractal for controlling the pitch of a melody. But I might use a simple random number generator constrained within limits for the dynamics. And I might have a set of durations (say 2, 3, 5, 8, and 13 pulses) which are selected between by a set of probabilities, so, for example a duration of 2 pulses occurs as much as all the other durations put together.

The problem that I usually face is: when I'm exploring a mathematical phenomenon with sound, what materials can express some aspect of the mathematical phenomenon without being simply and reductively a sonification. In a recent piece, for example, I made a series of sonification examples for my microtonality class. Later, I expanded these examples into a more finished piece, and in the process, I let my sonic imagination dominate the process, so that the sonification (where a demonstration of the qualities of the phenomenon is predominant) became music (where overall aesthetic criteria are predominant).

GM: I have trouble answering this question. If it means "is my piece a more or less faithful representation of the abstract process which is the input of the mapping?", the answer is generally no, since it seems that in most of the pieces I am thinking of, there is some aspect of the piece which is left up to my caprice. However, "Mandala 1086" and "Dry Rivers" were completely generated by programs without interference on my part (once the program was working to my satisfaction).

GS: I do not use consistent strategies. I use a mixture of the strictly procedural and the feral - letting both halves of the brain have a go!

DC: I'm not keen on 'technical' pieces, where the work seems to be purely about an artificial process, ignoring the sound & general aesthetic result. On the other hand, I like to minimise my role in the composition process, and so rely on technical or analytical approaches. If either the technical or aesthetic aspects are weak, I find that the work as a whole is shallow.

RA: I find I approach each situation as a new field of dreams. Sometimes the approach requires a more creative and holistic methodology but this usually gets pretty severely analysed, deconstructed, dissected etc.

PD: I do not often use an analytical approach; I prefer the creative approach with its greater freedom. For me, the music is the main goal. I will intervene in the process to achieve the aesthetic result I want. If I find that the mappings are not working and that I need to intervene continuously, I will revisit the mapping, not the overall structures as these are fundamental to the musical idea. Typically, by this stage I have a structure and data in place that I think is important for the composition, so the mappings will be manipulated to get the music, not the other way around.

PEB: This is difficult to answer because I'd say that I'd be categorised as using more creative, holistic mappings, but the approach is also always analytical. I don't think there is an actual separation between the 'aesthetic' and the 'analytical' other than a cultural enforced/assumed one.

JR: The movement of analysis and proposition is a seamless one, only formalised for the process of communication.

PXM: An analytical approach is important. I am always concerned to let a structure express itself as clearly as it can. The power of this kind of work is to denaturalise experience. It still has to look good, but

this is controlled by the choice of method, the rules of transformation.

4. Is the mapping component of your compositions something that you think might be perceptible by a listener, or of interest to them and why?

PM: There are cases in which I think it important for a listener to be able to perceive the procedures employed in a contextual sense. For example where I have developed software to generate spatio-temporal events, I have intended that the listener be able to perceive that the events have a context as a kind of spatial pattern in time. At other times I think that interesting results can be obtained without the listener necessarily being aware of the mechanisms that generated the music. An example of this is Takemitsu's use of hymns that are buried underneath a complex texture and occur so slowly that the listener cannot hear them. In Takemitsu's case the underlying structure of the hymn gives a consistency to the work without being overtly apparent.

GS: If a mapping component is likely to interest the audience, I like to make them play with it by occasionally doing the unexpected.

TK: This issue doesn't really concern me, ultimately the work is a final product and should stand on it's own merits as a piece of music. I guess this is the test of the structural integrity of my initial concept.

GM: Of course, it depends on the listener: if the listener is a composer of computer music, it is more likely to be of interest to the listener. In the case of "*New Stars*" I definitely wanted the listener to be able to relate the physical gestures that I made to the generated sounds.

CCB: gestures (such as 'sighs'), and the musically mapped equivalent, (descending and decrescendo musical objects) is surely of interest to the listener. It's part of the communication between composer and listener.

WB: If I want it to be heard, I'll generally do a talk and play session, trying to do this in as friendly a way as possible. But people can and do listen any way they want. A story from Kenneth Gaburo's composition seminar at the University of Iowa. A student was about to play an electronic composition. He said "It's based on the Fibonacci series, but you can't hear that..." Gaburo interrupted: "You mean it sounds like Mozart?" "No" the student says. "Well" Gaburo says, "if it doesn't sound like Mozart, or another kind of familiar music, if it sounds sort of unique, then maybe we ARE hearing the structure. We might not be able to RECOGNIZE it - like we recognize a Major chord, say, but the fact that it has it's own unique sound is very much a product of the system you chose to use."

GP: In my composition work, I believe the mappings are perceivable. The are central to the experience of the piece, in that I am looking to map material in such a way as to communicate the energetic/emotional core of the work. The are not academic/analytical in nature, so it is not a process of intellect through which they are perceived, but one of abstract experience. If you experience the energy of the piece, you experience the mappings central to the compositional 'craft' of the work. It seems to me that there is a difference between whether the mappings are perceivable by the listener, or whether they are of interest to the listener. The use of the term 'interest' implies an academic interest - the interest you as a composer may have, but not necessarily the focus of a non-composer listener.

DC: I would never use a process that was not relevant to what I hope a listener would perceive directly, or else use to frame their experience. My view is that if it cannot be experienced, then it is irrelevant to the work in its realised form.

PD: I think that at least some of them are perceptible to the listener, but there are typically many so I do not expect that all of them would be perceptible. I cannot say if it is of interest to the listener or not, perhaps it is to a few – it is important to me in as much as it leads to the musical concept that I'm trying to convey. Perhaps the mappings are somehow perceptible in the oppositions and tensions that I set up in a piece, although I use the mappings to try to make such tensions and explorations palpable, not to be perceived in themselves. I will often want the listener to perceive a sensuous, multi-dimensional surface or space, so in the manner that the mappings transmit this from the conception to the music and the listener, I think they may sometimes be perceptible.

PEB: They are perceptible even if this perception does not become conscious (ie. another mapping). The mapping is that which makes the work, it is in no way distinct from it. While it is impossible for the exact mapping process to be realised or known by

someone encountering it, the abstract set of relations acted out through the mappings are inescapable and fundamental to whatever interpretation is made.

JR: Ultimately, the product is autonomous of the processes of its production, although those processes may "render" the form in some way.

PXM: There are many other obligations to the viewer other than conceptual primacy. One would hope that the 'mapping component' is clearly present as an irreducible part of the 'sense' of the object.

VM: This is a difficult one to answer. It depends on who the viewer or user is. Generally the 'process' is of little or no importance outside the discipline and the knowledge it contributes to the discipline. At the end of the day the architecture is 'judged' along the criteria of architecture. The process is therefore only of interest for what it contributes to the questions of the discipline and I do not see it as an end in itself. I do though think that the process should be evident in the final work and if it is worth doing should have produced effects and conditions that are not possible without it.

5. Is the mapping component of algorithmic composition something that is pre-determined for you or is it part of a process of exploration?

PM: On a global level the answer is yes, there is usually a musical texture that I have in mind. However the specific algorithmic method employed is (as I mentioned above) obtained though experimentation.

GM: It often appears very early in the conception of a piece. (Though the elaborated melody process in the harpsichord piece came rather late.) Also the *details* of the mapping would be modified by exploration, but the basic idea of the mapping tends to be fixed at the start.

WB: More often than not, I'll get fascinated with a structure that would want to become sound, like a recurrent sequence, or a microtonal scale structure. The process of finding what sounds and structures are possible with that idea is one that is most usually interactive. Trying out one mapping, listening, changing things, etc. However, often I'll get to a point where I actually "don't like" the results I'm getting. In some cases with this, I will actually KEEP those results and listen to them for months, so that I can see if I can actually learn to hear (learn to like?) the results of applying the given structure to the chosen musical materials.

RA: This question presupposes a wide body of listeners. I have found that the EA music audience, at least from the more academic side we inhabit, has interest in "How You Did It". Other, inhabitants of the less academic side, have interest in the mechanics of the equipment used and pay finer, somewhat intuitive, attention to the sounds used. EG "Wicked kick, really cuts!" (I am not eau fait with the jargon). I think this process is reminiscent of that taken by rock bands, where the expression is ALL IMPORTANT. Back to the question. I guess Sometimes it may be of interest to a listener.

DC: Often pre-determined in a broad sense, but there is always a long process of exploration concerned with the details of implementation and presentation.

GP: As discussed previously - it comes from the material - it is a response to engaging with the sound material, and of working to express energetic states.

RA: It's part of the exploration, otherwise it would be no fun

GS: Mostly part of the exploration, though I wouldn't rule out it being pre-determined.

PD: Mapping is definitely part of the process of exploration. I tend to have a conception of a piece and the structures within it. I work at this level for some time and when the foundations are well laid and solid I start working on the mapping to generate the audible result I want.

PEB: Mapping is exploration and exploration always involves mapping.

JR: Professional practice has made mapping a baseline way of working, however in my own practice I do not view it as such.

PXM: I will take this as a question about selection of a structure, and the development of a method of concretization. The initial structure is chosen for its aesthetic potential, or conceptual importance. The method is the hard part and is nearly always derived by experimentation.

6. Do you use individual mapping strategies for individual parameters or is there reuse of mapping strategies or a global system? (Ie are they monoparametric or multiparametric?)

PM: There are obviously going to be approaches taken that are very non-specific applications of an

algorithm. For example my use of simulated neural networks was designed specifically as a means of generating spatio-temporal information (because that is exactly the kind of output from these systems). On the other hand I find myself constantly working with constrained random number generation to produce temporal events of one sort or another.

WB: I usually limit myself to one pattern per parameter, because I'm concerned with, in some sense, hearing the pattern I'm investigating. The idea of two processes adding to make a larger, more complex texture is appealing, but at least in my case, I would then have to listen to the overall sound produced and see how the two input processes could be tweaked to produce variations in the resulting mix-pattern.

RA: Integration is a vital philosophical approach in my composition/improvisation systems.

GP: This changes in accordance with the work being developed, and it's outcome. For instance, if the work is an installation piece, where music is generated through the use of real time video sensing, the mappings need to be multiparametric - that is, the characteristics of a single moving object must drive a number of algorithmic variables. I equate this to an instrumentalist who simultaneously controls a number of timbrel characteristics - for instance instrumentalists can create a number of differing timbres on any given pitch. They are skilled at creating differing amplitude envelopes, and combining these with a dynamic timbrel change to create an evolving sound that is appropriately contextualised within the composition. Even sophisticated electronic performance interfaces don't provide this level of nuance of control. At most they provide second order mappings.

PD: For each “line” of gestural or structural control I tend to use a single mapping, at least for the period of that compositional gesture, but this will be different from other mappings in the piece. I will also sometimes vary the scale of the mapping according to another parameter. For example, in a recent piece I varied the resynthesis “stickiness”, how easily a new harmonic would be triggered, with a gestural parameter, but the resynthesis range was controlled by another gestural parameter with another mapping. This is a somewhat unusual case for me. Also, at the same time, I needed to intervene for a period of a few seconds so that I achieved the required result at a critical point in the piece. This intervention actually was not with the mapping, although it could have been, but with the sensitivity of the resynthesis system at that moment. Of course, this is impossible when dealing with instruments so in such cases I need to adjust the mapping, or to intervene directly.

PEB: I always use multiparametric global mappings.

JR: Landscape is made of interactions, so separating factors can make it lose some of the richness of this interaction, despite the ease it gives to making them legible.

PXM: It varies. The problem with many interrelated parameters (parametric design) is that one designs a system for designing. In one way it doesn’t help with the problem of determining what the thing will be. There is a sense that the task is still in front of you - you need two sets of ideas, one to determine the constraints of the system, the second to determine an actual configuration. This is an important problem right now.

VM: Each circumstance is different, but there are overarching ideas and areas of investigation that tend to come up again and again, although the specific mapping or machine may be different

7. What elements do you control algorithmically in a composition and can you comment on the function and importance of these?

TK: Again this very much depends on the algorithm, there are usually many mapping permutations applied in different (or the same) contexts.

PM: I usually prefer to work with real-time applications and I feel that in these cases there should be a high level of user gratification (that is if you make a change to a parameter there should be a reasonable level of “response”. For this reason I find that I tend to prefer systems in which the algorithmic process employed is governed by the user interface. An example of this is my implementation of a flocking algorithm. The user controls the direction of flock in two dimensions while the algorithm generates values that are distributed around the control curve.

CCB: They serve the functions as station points, as a short reminder of returning to material heard previously in the work, and as sonic symbols

(although rarely). Their Importance is that they are often very significant and of great importance.

WB: It depends on the piece. Although, since I come from a middle of the road, easy-listening musical background (jazz accordion, etc), and NOT from the Classical Music Tradition, I often find that I have no desire whatever for dynamic contrast in my music, so often the dynamics are the least shaped element of my music. At one time I thought this was a lack, and something that I needed to work on, but now that I'm feeling more connected with my "lounge" roots, and am more and more alienated from anything having anything to do with the whole world of European Classical Music, this sense of even, non contrasted dynamics is something I'm now consciously and enthusiastically embracing.

GS: My preference is to control as few elements as possible myself with the system controlling as few as possible. That way the contest is about even with no match fixing.

GM: I usually control only some elements, although I would like to do some more purely algorithmic compositions, in which all or most elements would be controlled algorithmically.

RA: Again, the process is fluid from composition to composition.

GP: I do not use algorithms to establish/set or control temporal form. Algorithms are used only for sound production and therefore for orchestration.

PD: I try to control as many elements as possible, but that control may be more of a general direction, so perhaps I am not such a control freak. I do use a lot of controlled randomness in both structures and gestures and the mappings.

PEB: Control is never properly situated in elements, even though some may have a more primary constraining or defining function (such as a latex mould which provides a deformable shape that links all casts into the series). Control occurs through elements rather than the elements being controlled. As such, control occurs loosely but in an encompassing, overall manner through qualities and tendencies that will respond to the moment of process.

PXM: My previous comments about resolution are valid here. One approach would be to take as much control as possible initially, then work out where to relax control in order to make the thing work.

8. Is the mapping consistent within these elements or not?

PM: Within a given approach yes, my mappings are consistent, as it would be difficult to achieve user gratification if there was not a high level of consistency.

WB: The previous answer is relevant here. There is the example of Dary John Mizelle, however, who likes to pile simple process upon simple process upon simple process until a chaos results...so if I'm doing that, I'll try to be as inconsistent as possible. I'm also attracted to the Herbert Brun-Felix Werder argument that since logic has so spectacularly failed us, to make music which reflects logical structures is, at best, simply to be indulging in wishful thinking and nostalgia, and, at worst, to be promoting a worldview that has proven disastrous to humanity in the past, and will most likely continue to do so.

GM: I'm not sure I understand this question. I took it to mean that with pitch (for example), I would generally use one mapping throughout the piece or section of a piece, rather than (capriciously) using several different mappings. If I had a systematic way of using several mappings, it seem likely that I would view this as a single mapping.

DC: I tend to be totally consistent.

RA: I'm mostly consistent, this allows for a degree of predictability

PEB: If I replace what you call 'elements' with 'qualities and tendencies' then the mapping in which they are involved produces the consistency of the work.

JR: Representation is complicit with seeing, so its not useful to force the world into uniform boxes if one wishes ones work to have an interpretive role.

PXM: It depends on the piece, and what is salient in the structure one is interested in. Nonlinearity can make it harder to sense the strategy within the work - it is not that interesting in its own right.

9. Are the mapping schemas you use mostly linear mappings or non-linear in some fashion, and why?

TK: This very much depends of the perceptual construct of the parameter. Also some algorithms are by their nature non-linear and require being treated as such.

PM: Mainly for the reasons that I outlined in question 2. I tend to think that linear mapping involves arbitrary decisions whereas approaches in which a user interacts with a generative system of some kind feel more organic or at least more integrated with my musical thought.

GM: In the strict mathematical sense the notion of linear mapping only applies when the input and the output quantities have a notion of addition and a notion of (scalar) multiplication. I think you have a more general notion in mind, but it seems that at least a notion of ordering is involved: "more" input, in some sense (eg. larger numbers) gives "more" output (eg. louder, higher). If you are dealing with things like timbres or melodies, I think the notion probably does not make sense.

GS: Non-linear are the most interesting but also the most confusing. Using linear and non-linear makes for the best mix of intelligible (and potentially boring) and interesting (and bewildering).

DC: Linear I guess. This is because the sonic implications of the mapping are complex and subtle, so it does not make sense to add gratuitous complexity. I always go for simplicity and elegance but also naturalness.

RA: I use mostly nonlinear mappings, I like the idea of diverse interaction

GP: The mappings are in general non-linear. I find that rates of change are important, and tend towards logarithmic characteristics, although I find I often end up writing the functions or wave tables that are being used to interpolate the input controller onto the variable, because I inevitably find that I wish for some variation in the characteristic of the standard logarithmic distribution. I have tended to feel (in a rather uninformed and un-researched way) that logarithmic distributions (scaling) is in line with human perception (al la amplitudes -> perceived loudness in decibels). As I mentioned previously, I approach composition intuitively, so the mappings that are used in my composition are largely arbitrary in response to material I am hearing. These mappings are therefore generally not linear - they change in accordance with the momentary material of the composition.

PD: I use a combination of linear and non-linear mappings, depending on the data and the musical parameters to be output. It depends on the structural model or gesture or data that I'm working with. I do this because I think it sounds much more interesting in the music.

PEB: Generally there is a broad envelope for any project that is quite linear - but this is always forced (you decide there's a beginning and an end to a process and locate them in some relatively arbitrary moment). Otherwise, within these brackets, it's all non-linear.

JR: Design schemas are by linear by convention, relying on a narrative sequence to communicate things, however this does not actually suit the nature of landscape. Using the best means available.

VM: I am not sure I understand the question. But the mappings used are selected in order to achieve conditions & levels of complexity that could not have been considered or designed without them. Generally they are complex and abstract systems.

10. Can mapping be considered a composition technique in itself?

TK: I've never separated mapping from composition, the two are so intimately bound for me. Its the same for me as selecting a serial matrix, harmonic sequence or constructing and treating a melody in a traditional sense. My understanding of traditional constructs mostly informs my mapping although there are times when I feel hindered by this perspective when trying to find solutions to difficult mapping problems.

PM: Once again it depends on the level that you are talking about. I suppose that a total mapping approach would not be very interesting to me I would not be comfortable with calling it a means of global compositional technique. On a small event scale however the answer has to be yes. If you use a method to generate a sound event then it has to be a compositional gesture.

WB: For me, mapping is trying out different materials with different patterns. The surface of the music is often determined by the materials to the detriment of the patterns. (IE- any pattern applied to a C major scale played by flutes and harps with a slow tempo and lots of reverb will probably sound like new age music, even if you're pattern is 100% equal weighted random). So if one is interested in using or exposing different aspects of patterns, careful selection of materials is often essential.

RA: Absolutely, mapping can be a compositional technique in itself. In our PoMo (PoPoMo) age

taking from one to generate/create/influence/etc the other is de rigeur.

GP: yes, mapping can be used as a composition technique in itself - it often is - however, I don't generally find the outcome from a process driven piece to be aurally/aesthetically pleasing. I find Xenakis' comments of the use of algorithms to develop material which the composer then makes selections from, and alterations to as part of the composition process, to be enlightening. The small momentary nuance of the composition need in my view to be made in a conscious way by the composer - this is I think the craft of composition. A strategy for varying mappings in accordance with compositional states is another way of looking at this issue. Evolving mapping approaches form a more interesting technique.

PD: I think that mapping is only part of the technique of algorithmic composition. Certainly, someone could take some predefined data, such as weather data, and use mostly mapping techniques to generate a piece of music. And it might be wonderful music. However, I tend to think very abstractly about most of my music and thus I like to construct the compositional gestures and data from the outset, and this is also part of composition.

PEB: In the way I understand mapping, architectural design cannot occur without mapping.

JR: It can, but should not be - referencing something that's in the real world begs one to question why it should be separated from its actual existence.

PXM: Yes – mapping alone is quite enough in terms of the sense of the project. However it is difficult to find a strategy powerful enough to inform all decisions, especially in a resolved design - the idea of grey - 'not part of central design idea' is not really adequate, because it immediately creates a background against which the 'interesting' parts can be read. Grey is a kind of decorous strategy that suggests that a designed thing is really just a frippery existing in a background world we can take to be constant and true. I have a profound dislike for any design that enforces the normative through its own exceptional qualities - this is actually nearly all design.

VM: Mapping is a part of a larger process. It would be far too simplistic in architecture to consider the design the built mapping.